Complier Design - Assessment Tool

Q1 The Address code for the if-else statement
Given statement:

if ((a<b) and ((c<d) or (a>d))) then
z = x + y * z
else
z = z+1

Sol
step1:

The condition is
(a<b) AND ((c<d) OR (a>d))
a<b
c<d
a>d

(a<b) AND [(c<d) OR (a>d)] is structure

step2:

L1, L2, L3, L4, L5

step3:
L1:
if a<b goto L2
goto L5

L2 :
if c<d goto L4
goto L3

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L3:
if a > d goto L4
goto L5

Step 4:

L4:
t1 = y * z
t2 = x + t1
z = t2
goto L6

Steps:

L5:
t3 = z + 1
z = t3

Final TAC for Q1

L1:
if a < b goto L2
goto L5

L2:
if c < d goto L4
goto L5

L3:
if a > d goto L4
goto L5

L4:
t1 = y * z
t2 = x + t1
goto L6

Q2 Three Address code using backpatching

(a<b) and (c<d) or (e<f)

Sol step 1

[(a<b) AND (c<d)] OR (e<f)

step 2

1. if a<b goto _
2. goto _
3. if c<d goto _
4. goto _
5. if e<f goto _
6. goto _

step 3.

(a<b) AND (c<d), if a<b is true

Back patch statement 1 to statement 3

1. if a<b goto 3

step 4:

Back patch statement 2 to 5

Back patch statement 4 to 5

steps:

1. if a<b goto 3
2. goto 5
3. if c<d goto 7
4. goto 5
5. if e<f goto 7
6. goto 8
7. TRUE
8. False

Q3 TAC for Boolean Expression using Given Productions
B->B1 || B2, B->B1 && B2, B->!B1, B->E1 relop E2, B->true
B->false

Step 1:

[(a>b) OR (c<d)] OR (e<f)

Step 2

L1:
if a>b goto LTRUE
goto L2

Step 3

L2
if e<f goto LTRUE
goto L3

Step 4

L3:
if e<f goto LTRUE
goto LFalse

Final

L1:

if a > b goto LTRUE

goto L2

L2:

if c < d goto LTRUE

goto L3

L3:

if e < f goto LTRUE

goto LFalse

LTRUE

goto LEND

LFALSE

LEND

Q4 TAC for the while - if - else program

while (A < C and B > D) do

if A = 1 then

C = C + 1

else

while A <= 0 do

A = A + B

Step1:

L1:

if A < C goto L2

goto LEND

L2:

if B > D goto L3

goto LEND

step 2:

L3:

if A==1 goto L4

goto L5

Step 3

L4:

t1=C+1

C=t1

goto L1

Step 4

L5:

if A<=B ~~goto~~ goto L6

goto L1

step5

L6:

t2 = A+B

A = t2

goto L5